CSE 270 Sports Analytics

PSS3: Competitive Balance

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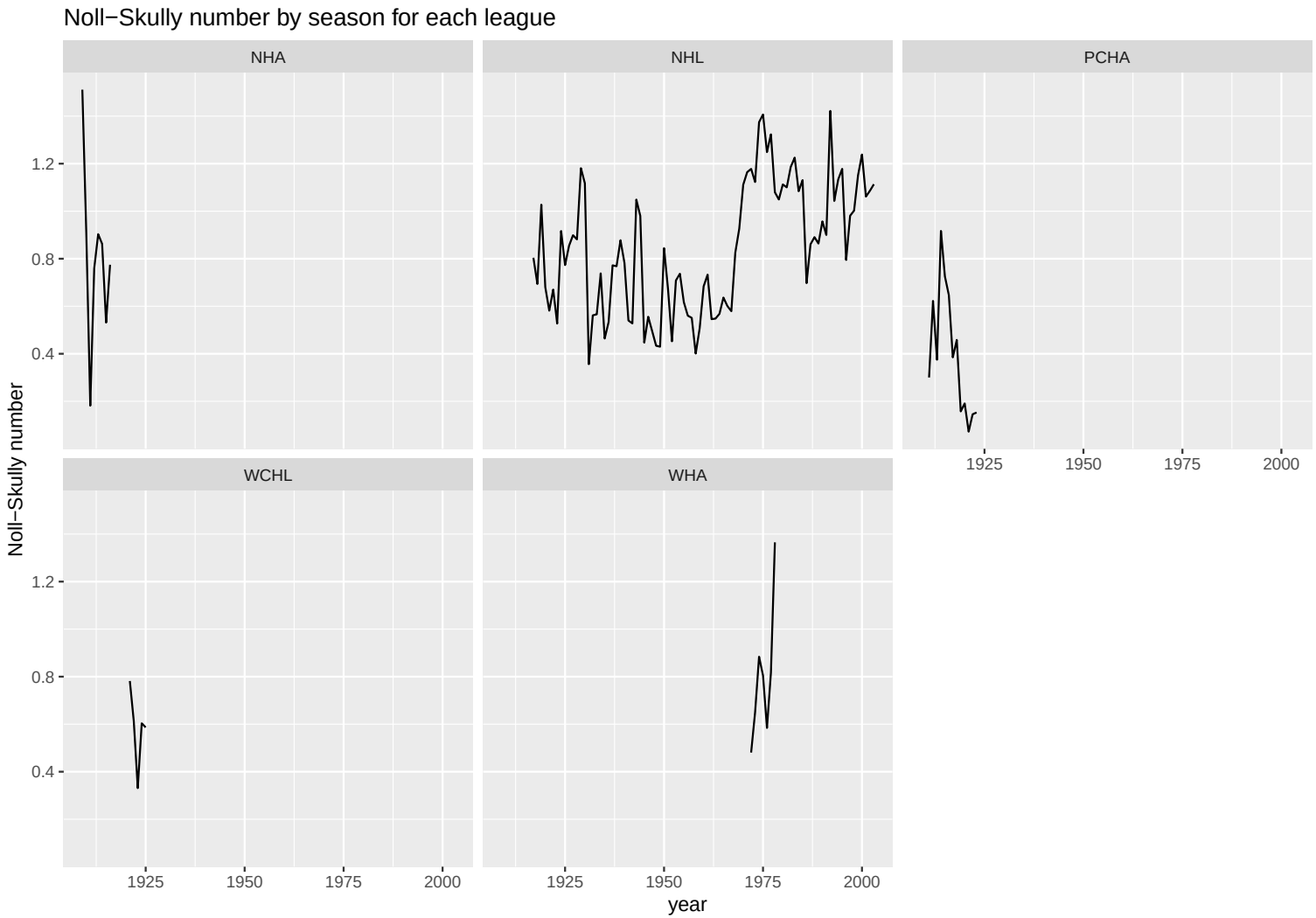
Competitive Balance

Noll Scully number

```
nhl_standings <- read.csv("./Teams.csv") %>%
  group_by(lgID) %>%
  arrange(year,rank) %>%
  ungroup() %>%
  filter(year < 2005)
nhl_standings$WR <- (nhl_standings$W + nhl_standings$T/2) /
  (nhl_standings$W + nhl_standings$T + nhl_standings$L)
noll_scully <- nhl_standings %>%
  group_by(year, lgID) %>%
  summarize(NSN = sd(WR) / (0.5 / sqrt(n())) )
```

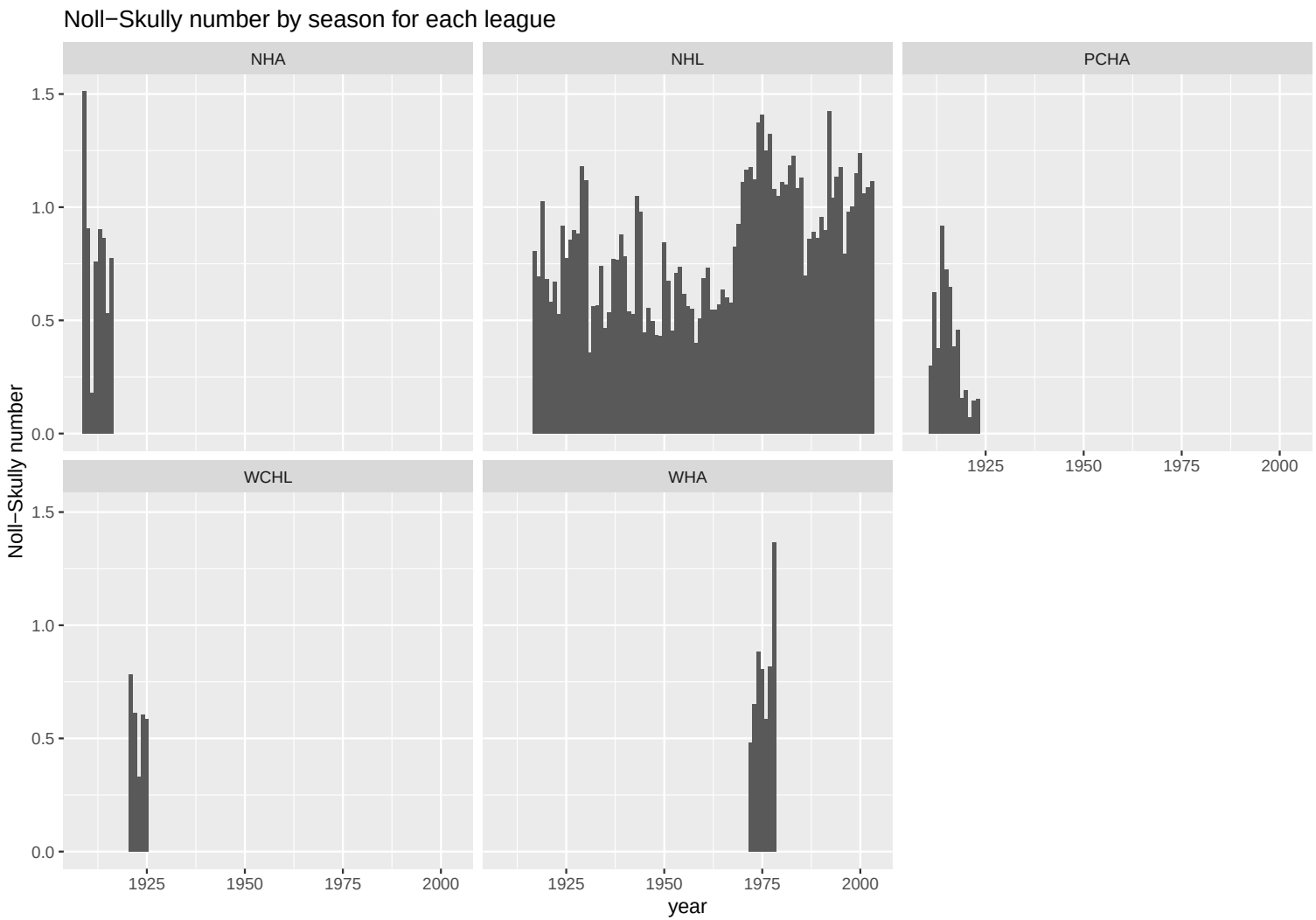
Noll Scully number

```
ggplot(noll_scully,aes(x = year, y = NSN)) +
  geom_line() +
  ggtitle("Noll-Skully number by season for each league") +
  ylab("Noll-Skully number") +
  facet_wrap(~lgID)
```



Noll Scully number

```
ggplot(noll_scully,aes(x = year, y = NSN)) +
  geom_bar(stat="identity") +
  ggtitle("Noll-Skully number by season for each league") +
  ylab("Noll-Skully number") +
  facet_wrap(~lgID)
```



Individual performance

Individual performance

```
fifa23 <- read.csv("./Fifa 23 Players Data.csv")

fifa23 <- fifa23 %>%
  select(Best.Position, Preferred.Foot, Skill.Moves, Weak.Foot.Rating,
         Attacking.Work.Rate, Defensive.Work.Rate, Shooting.Total, Defending.Total,
         Passing.Total, Pace.Total, Height.in.cm., Weight.in.kg.,
         Goalkeeper.Reflexes, Goalkeeper.Diving, )
```

One hot encoding for categorical variables

```
fifa23 <- fifa23 %>%
  mutate(Preferred.Foot = as.factor(Preferred.Foot),
         Skill.Moves = as.factor(Skill.Moves),
         Weak.Foot.Rating = as.factor(Weak.Foot.Rating),
         Skill.Moves = as.factor(Skill.Moves),
         Attacking.Work.Rate = as.factor(Attacking.Work.Rate),
         Defensive.Work.Rate = as.factor(Defensive.Work.Rate)
         )
new_fifa <- one_hot(as.data.table(fifa23))
```

Fitting a model

```
fifa_players_train <- new_fifa %>%
  sample_frac(0.80)

fifa_players_test <- new_fifa %>%
  anti_join(fifa_players_train)

mod.nnet <- nnet::multinom(
    Best.Position ~ .,
    data = fifa_players_train
)
```

```
## # weights:  420 (378 variable)
## initial  value 40163.092533
## iter  10 value 27639.255336
## iter  20 value 20533.002226
## iter  30 value 18704.005250
## iter  40 value 17641.483369
## iter  50 value 17289.826877
## iter  60 value 17132.114650
## iter  70 value 17017.478199
## iter  80 value 16864.119361
## iter  90 value 16713.629754
## iter 100 value 16331.357486
## final  value 16331.357486
```

Predicting on test set

```
fifa_players_test$PrdPos <- predict(mod.nnet, newdata = fifa_players_test)
mean(fifa_players_test$PrdPos == fifa_players_test$Best.Position)
```

```
## [1] 0.6369583
```